THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 02

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** mass and velocity	symbols stand for quantities
2	**(a)** direction — choosing models and the right questions	exploration needs direction
3	**(b)** they are too small	size less than about 1 nm
4	**(c)** simple distillation	e.g. acetone and water
5	**(c)** Atoms cannot be created or destroyed in a chemical reaction	indivisible and conserved in reactions
6	**(b)** high melting and boiling points	strong attraction between ions
7	**(b)** the stability of the atom	an orbiting electron would lose energy
8	**(c)** zero	returns to start
9	**(c)** 2πr/T	circumference ÷ time
10	**(d)** 2.5 m s^{-2} east	(8 − 3)/2
11	**(a)** 6 m s^{-1}	a = F/m, v = at
12	**(c)** 50 J	½mv^2
13	**(b)** 340 m s^{-1}	v = ν λ
14	**(d)** ribosomes	attached to the RER or free
15	**(b)** cell	cell theory
16	**(c)** cartilage	flexible
17	**(a)** species	can interbreed and produce fertile offspring
18	**(d)** reptiles, birds and mammals	inside the female body
19	**(a)** penicillin	from Penicillium
20	**(a)** CFCs	refrigerators, aerosols

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	Re-examine the assumptions, the model or the measurements instead of rejecting evidence (2).	2
22	By sharing of electron pairs between atoms to complete the octet/duplet (1); e.g. H_2, Cl_2 or HCl (1).	1+1
23	Tangent to the circle at the point (2).	2
24	W = mg = 30 × 10 = 300 N (2).	2
25	It contains DNA with genes controlling cell activities (2).	2
26	Its dead cells have thick, lignified walls (2).	2
27	Provides oxygen and nutrients to the foetus and removes its wastes (2).	2
28	Uneven heating creates pressure differences; air moves from high to low pressure (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Evaporation removes the solvent completely and may leave impurities (1.5); crystallization uses cooling of a hot saturated solution and gives pure crystals (1.5).	1½+1½
30	Natural chlorine is a mixture of isotopes in unequal proportions (1); the weighted average takes abundances into account (1); no single atom has mass 35.5 u (1).	1+1+1
31	15 × 2 = 30 × d (2); d = 1 m (1).	2+1
32	Bats emit ultrasonic waves (1); sense the echoes from objects (1); judge the distance and size of obstacles and prey (1).	1+1+1
33	Plasma, red cells, white cells, platelets (1.5); transport of oxygen, nutrients, wastes and defence (1.5).	1½+1½

34	It has many endemic species under threat (1.5); sacred groves are traditionally protected forest patches that conserve local species (1.5).	1½+1½
35	carbon disulfide; dinitrogen tetroxide; phosphorus trichloride; hydrogen sulfide; ammonia (2). Prefixes mono, di, tri… show numbers of atoms; no prefix on hydrogen or mono for the first element (1).	2+1
	OR: Na (2,8,1) loses 1 electron to become Na^+ (2,8) (1); Cl (2,8,7) gains 1 electron to become Cl^- (2,8,8) (1); opposite charges attract forming an ionic bond (1).	1+1+1
36	India-led initiative encouraging mindful, eco-friendly lifestyle choices by individuals and communities (3).	3
	OR: Less transpiration and less infiltration, more runoff and erosion (1.5); less rainfall and soil loss (1.5).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $CaCl_2$, Al_2O_3, Na_2SO_4, MgO, NH_3 (2.5). (ii) Cross-over of valencies / criss-cross method (1). (iii) carbon monoxide, carbon dioxide, carbon disulfide, dinitrogen tetroxide, phosphorus trichloride (1.5).	2½+1+1½
	OR: (i) A neutral entity of more than one atom that can exist independently (1). (ii) Element: H_2, O_2, Cl_2; compound: H_2O, HCl, CO_2 (3). (iii) Its valence shell is already full (1).	1+3+1
38	(i) Mass is fixed, weight depends on g (2). (ii) 600 N (1.5). (iii) 96 N (1.5).	2+1½+1½
	OR: (i) Weight downwards, her upward force (1). (ii) 294 N (1). (iii) 294 N upwards (1). (iv) Zero (2).	1+1+1+2
39	(i) Biconcave, no nucleus, full of haemoglobin (2). (ii) Nerve cell, muscle cell or root hair cell with its function (2). (iii) About 120 days (1).	2+2+1
	OR: (i) 4 mm ÷ 20 = 0.2 mm = 200 µm (1). (ii) 100X (1). (iii) 200 µm × 100 = 20 000 µm = 20 mm (2). (iv) Resolution of the lenses (1).	1+1+2+1
40	(i) Testes, sperm duct, urethra; ovaries, oviducts, uterus, vagina (3). (ii) Testes: sperms and hormones; ovaries: eggs and hormones (2).	3+2
	OR: (i) Puberty (1). (ii) Hormones (1). (iii) Onset of menstruation, development of breasts (1). (iv) Supports rapid growth (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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